

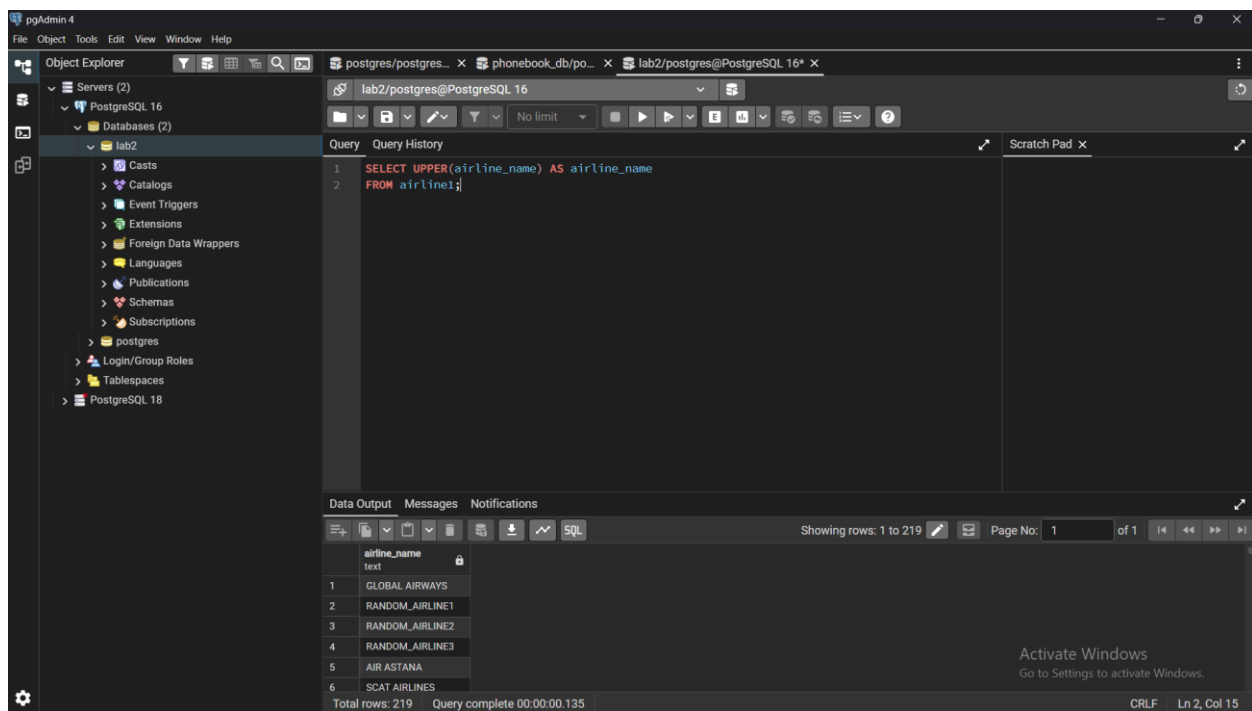
Laboratory work 4

We continue to work with the database from the previous laboratory works.

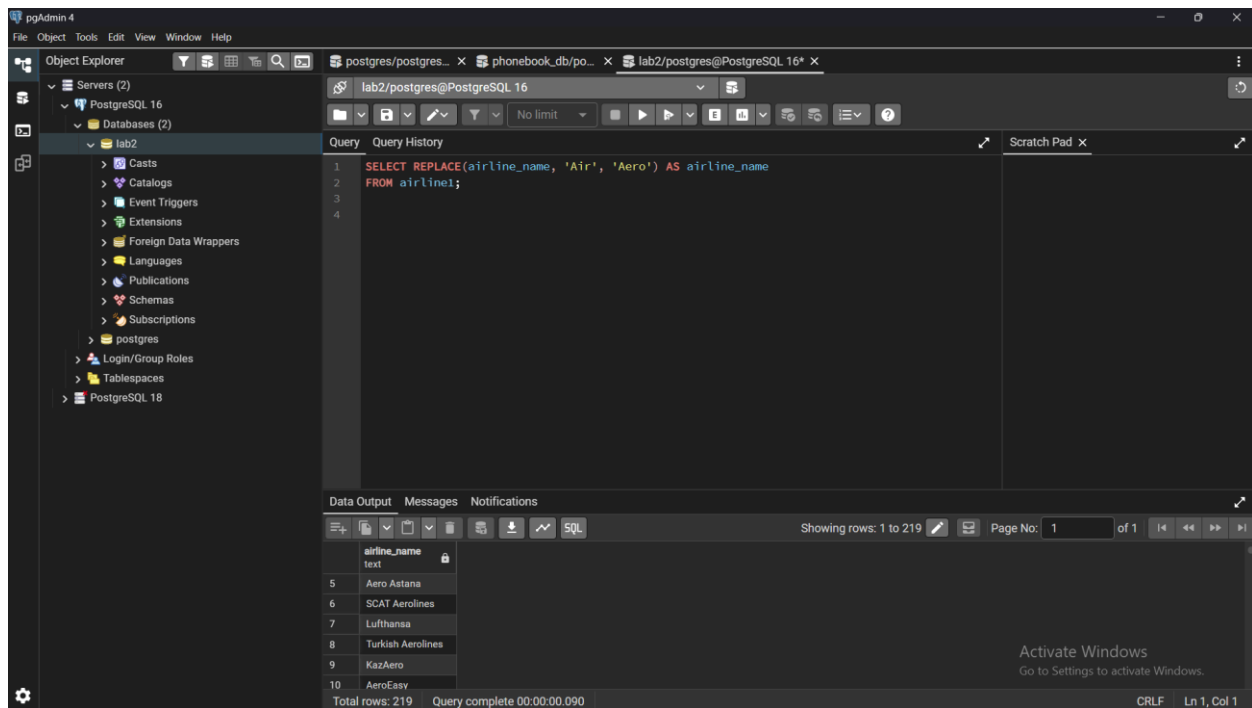
Take a full-page screenshot that covers the code and results of each task.

Tasks:

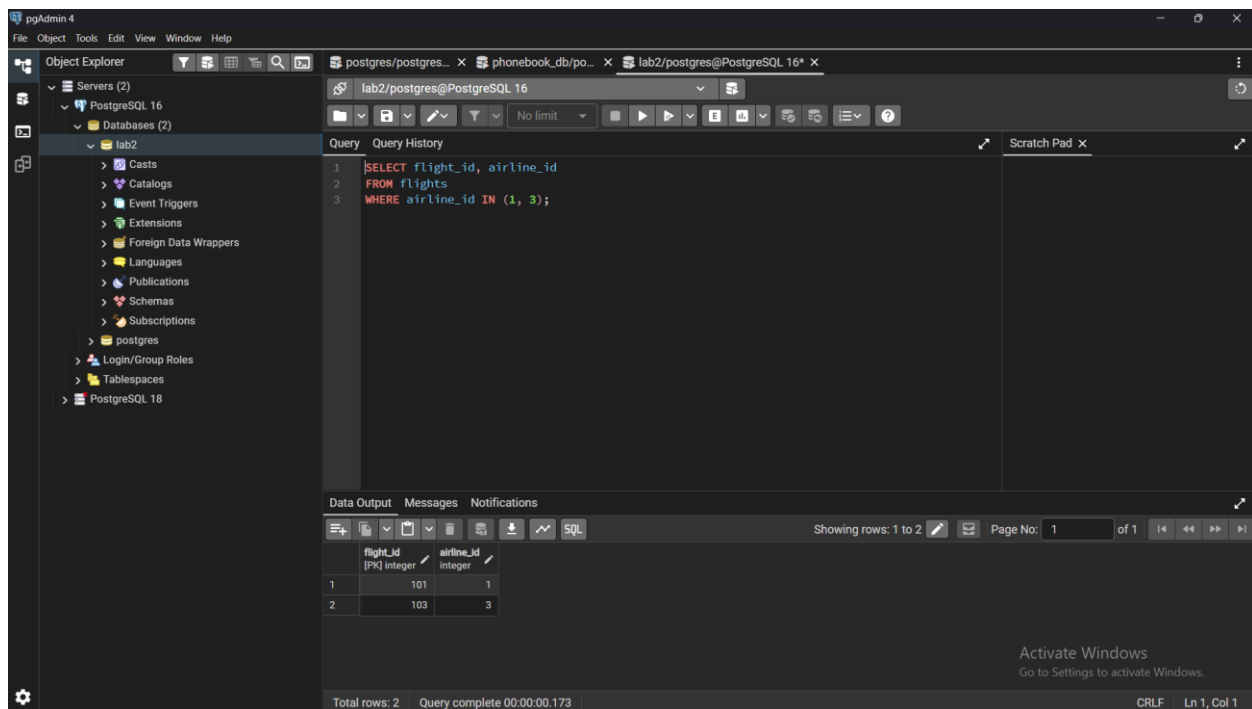
1. Retrieve all airline names in uppercase.



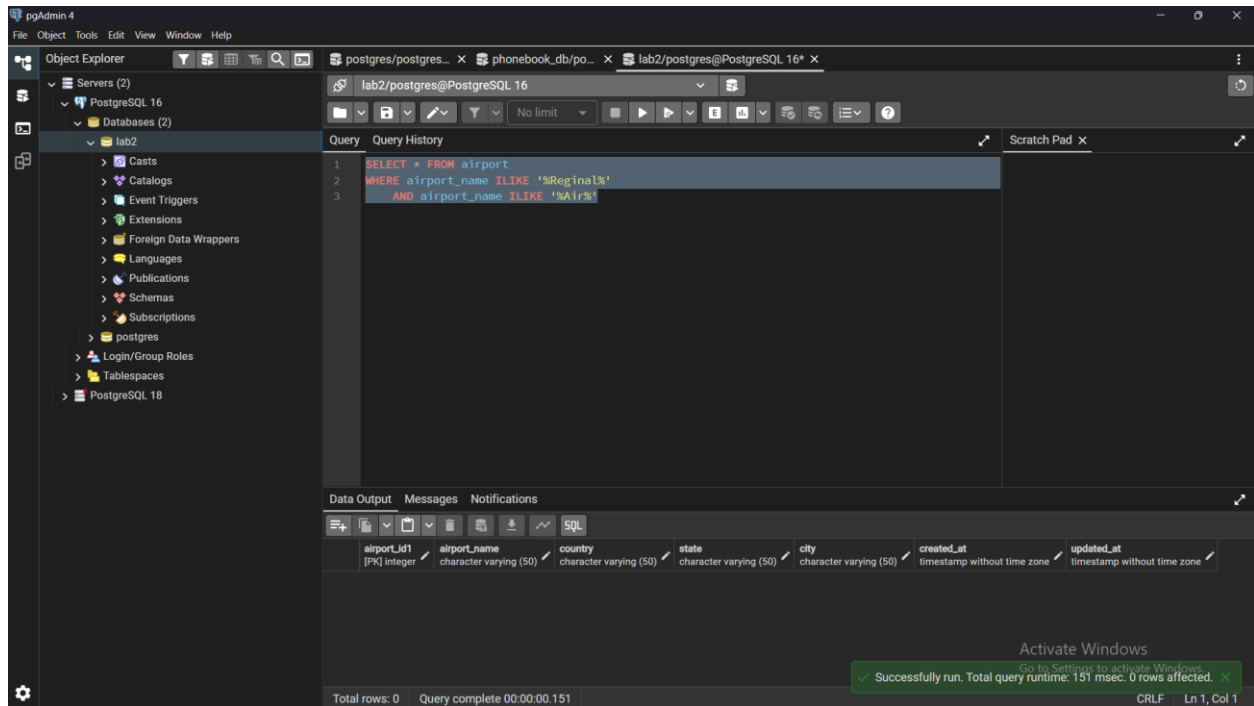
2. Replace any occurrence of the word "Air" in airline names with "Aero".



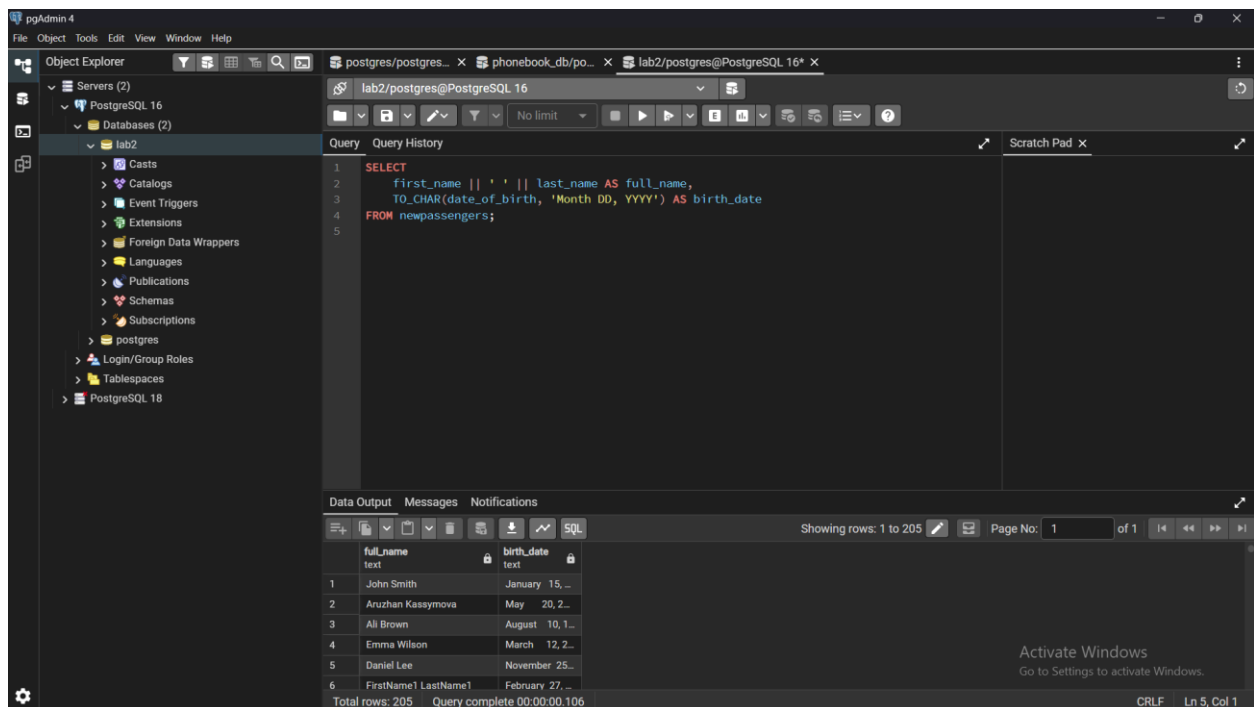
3. Find all flight numbers that coordinates with both airline 1 and airline 2.



4. Retrieve airports that contain the word "Reginal" and "Air" in their names.



- Retrieve passenger names and format their birth dates as 'Month DD, YYYY'.



- Find flight numbers that have been delayed based on the actual arrival time.

The screenshot shows pgAdmin 4 with a query window open. The query is:

```
1 SELECT flight_id, sch_arrival_time, act_arrival_time FROM flights
2 WHERE act_arrival_time > sch_arrival_time;
```

The results are displayed in the Data Output pane:

flight_id [PK] integer	sch_arrival_time timestamp without time zone	act_arrival_time timestamp without time zone
1	2026-09-21 12:00:00	2026-09-21 12:05:00
2	2026-09-23 18:00:00	2026-09-23 18:05:00
3	2026-09-24 15:00:00	2026-09-24 15:10:00
4	2026-09-25 20:00:00	2026-09-25 20:10:00

Total rows: 4 Query complete 00:00:00.169

- Find flights that arrived late according to their actual arrival time compared to the scheduled arrival time.

The screenshot shows pgAdmin 4 with a query window open. The query is:

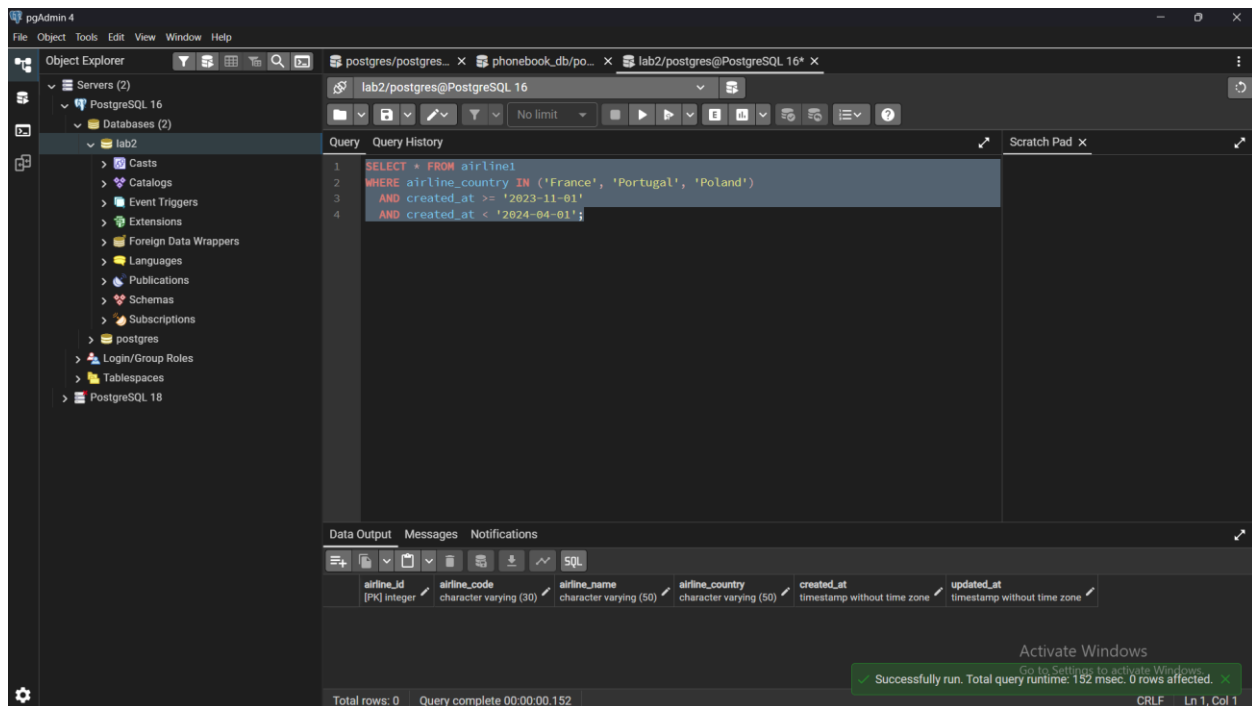
```
1 SELECT flight_id, sch_arrival_time, act_arrival_time FROM flights
2 WHERE act_arrival_time > sch_arrival_time;
3
```

The results are displayed in the Data Output pane:

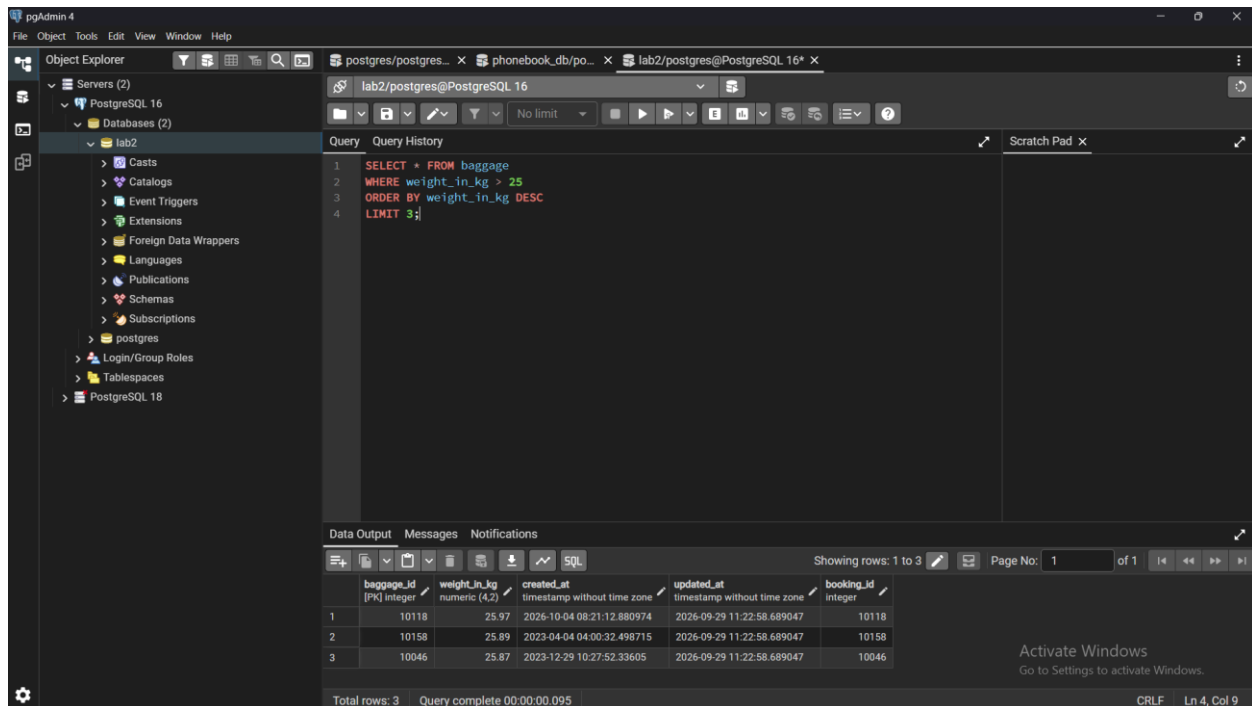
flight_id [PK] integer	sch_arrival_time timestamp without time zone	act_arrival_time timestamp without time zone
1	2026-09-21 12:00:00	2026-09-21 12:05:00
2	10001	2026-05-24 05:07:00.076727
3	2026-09-23 18:00:00	2026-09-23 18:05:00
4	2026-09-24 15:00:00	2026-09-24 15:10:00

Total rows: 114 Query complete 00:00:00.196

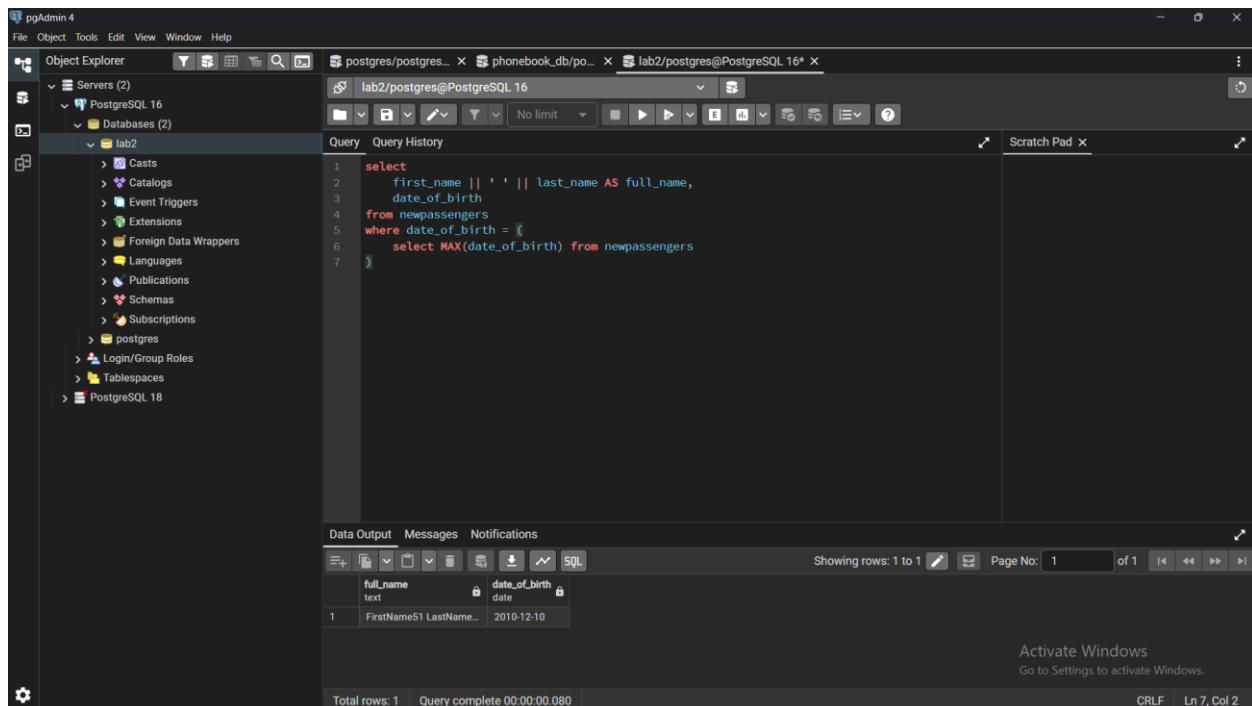
- Show airlines from any of: ('France','Portugal','Poland') created between '2023-11-01' and '2024-03-31'.



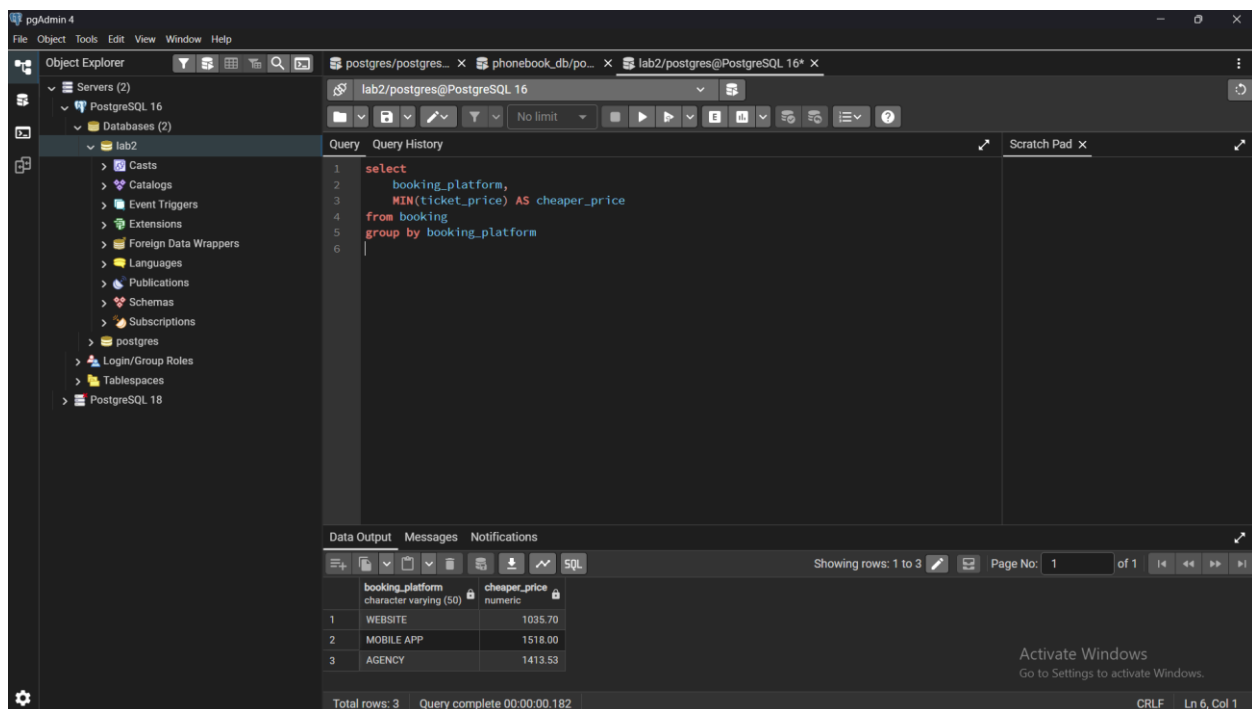
9. Find top3 overweighted baggage with more than 25kg.



10. Find the youngest passengers' full name.



11. Find the cheapest booking price on each booking platform.



12. Return airlines whose airline_code contains a digit.

pgAdmin 4

Object Explorer

Servers (2)

- PostgreSQL 16
 - Databases (2)
 - lab2
 - Cast
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas
 - Subscriptions
 - postgres
 - Login/Group Roles
 - Tablespaces

lab2/postgres@PostgreSQL 16

Query

```

1 select * from airline1
2 where airline_code ~ '[0-9]';

```

Query History

Scratch Pad

Data Output Messages Notifications

Showing rows: 1 to 209

Page No: 1 of 1

airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	1201	A1	Random_airline1	Country1	2026-09-29 10:57:52.674363
2	1202	A2	Random_airline2	Country2	2026-09-29 10:57:52.674363
3	1203	A3	Random_airline3	Country3	2026-09-29 10:57:52.674363
4	1011	A11	Random_airline11	COUNTRY11	2026-09-28 12:02:23.630343

Total rows: 209 Query complete 00:00:00.184

CRLF Ln 1, Col 1

13. List the top 5 most recently created airlines.

pgAdmin 4

Object Explorer

Servers (2)

- PostgreSQL 16
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lab2/postgres@PostgreSQL 16

Query

```

1 select * from airline1
2 order by created_at desc
3 limit 5;

```

Query History

Scratch Pad

Data Output Messages Notifications

Showing rows: 1 to 5

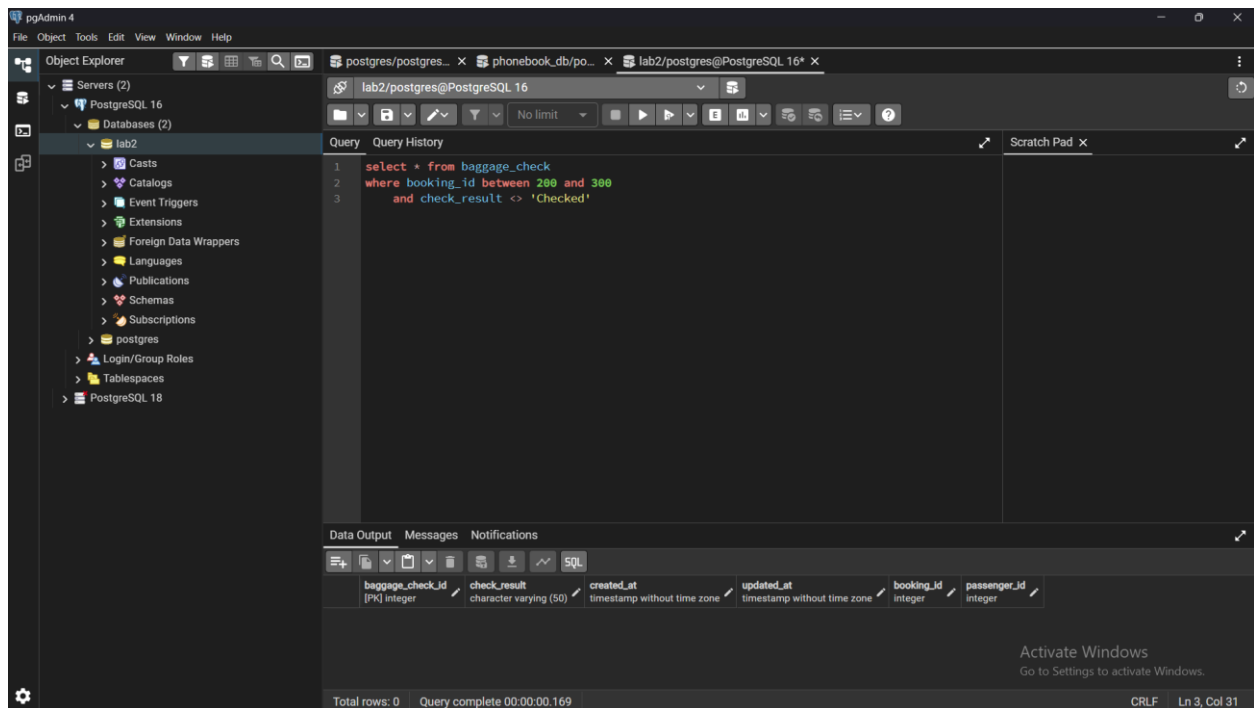
Page No: 1 of 1

airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	1204	A4	Random_airline4	Country4	2026-09-29 10:57:52.674363
2	1202	A2	Random_airline2	Country2	2026-09-29 10:57:52.674363
3	1203	A3	Random_airline3	Country3	2026-09-29 10:57:52.674363
4	1201	A1	Random_airline1	Country1	2026-09-29 10:57:52.674363

Total rows: 5 Query complete 00:00:00.406

CRLF Ln 1, Col 1

14. Return all rows where booking_id is between 200 and 300 inclusive and check_result <> 'Checked'.



15. Baggage checks where `update_at` is in the same month as `created_at` but occurs earlier than `created_at`.

